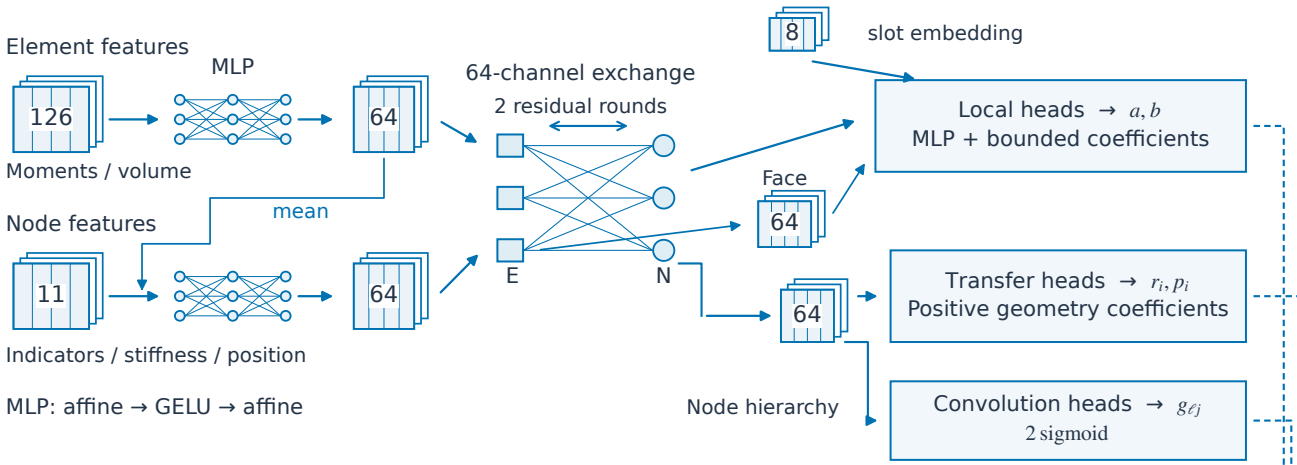
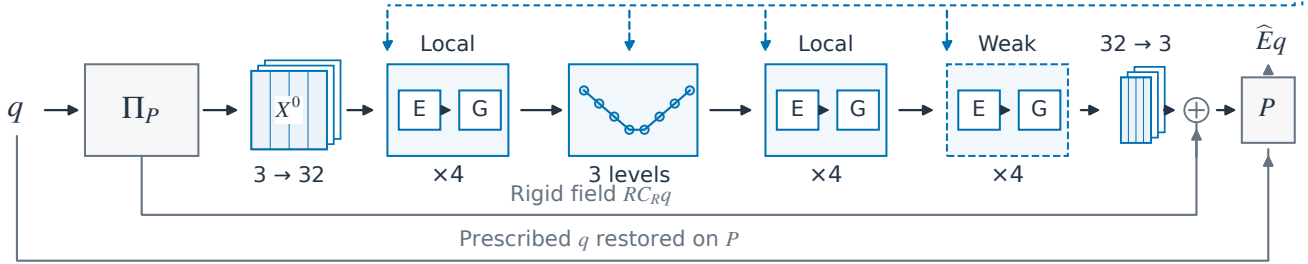


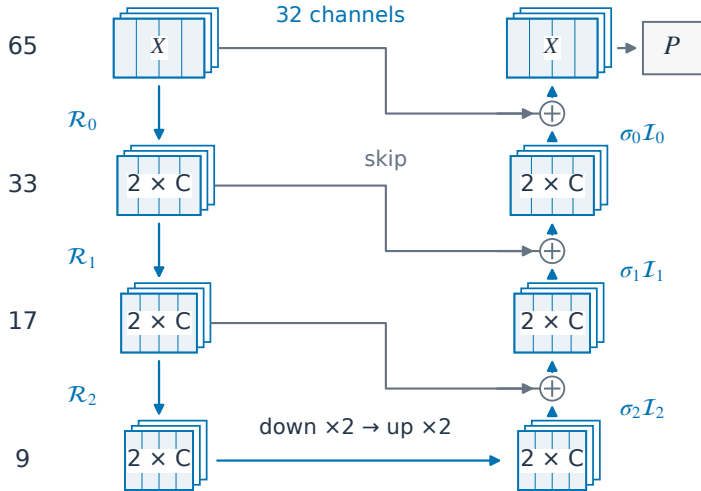
(a) Geometry branch



(b) Displacement branch: linear in q for fixed geometry



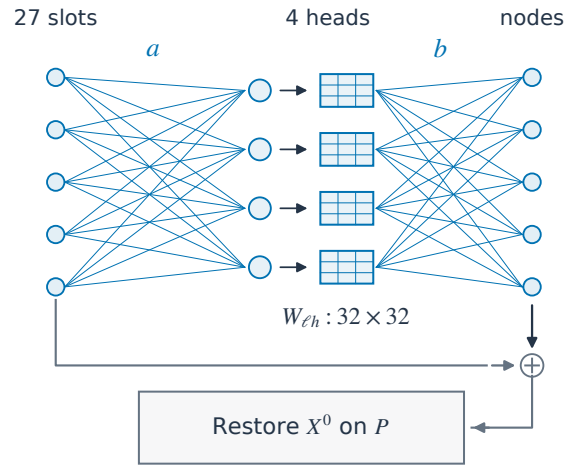
(c) Multiscale displacement propagation



$\mathcal{R}$: restriction $\mathcal{I}$: prolongation

$C: X \leftarrow X + g \odot \text{Conv}_{3 \times 3 \times 3}(X)$

(d) Local interaction



E : 27 Q2 nodes

G : 18 owner + 9 neighbour nodes

Training: update network parameters θ through both branches. Deployment: fix θ ; cache geometry coefficients.